IdiomX: A Multilingual Benchmark for Idiom Detection and Cross-Lingual Retrieval

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Abstract

Idiomatic expressions remain a fundamental challenge in natural language processing due to their non-compositional semantics, strong contextual dependence, and difficulty in cross-lingual alignment. Existing idiom datasets are often limited in scale, contextual diversity, or multilingual coverage, restricting their effectiveness for modern language models.

In this paper, we introduce IdiomX, a large-scale multilingual (English–Arabic–French) benchmark for idiom understanding and cross-lingual semantic retrieval. IdiomX is constructed through a reproducible multi-stage pipeline that integrates lexical resource extraction, large-scale cleaning and normalization, controlled large language model (LLM) enrichment, and structured validation. The pipeline further incorporates semantic consistency checks, deduplication, and leakage-safe splitting to ensure high-quality data and reliable evaluation.

The resulting dataset contains over 190K contextualized examples covering more than 12K idioms, with aligned English, Arabic, and French semantic representations, idiomatic and literal usage labels, and rich linguistic metadata.

We define a unified benchmark with three tasks: idiom detection, context-to-idiom retrieval, and Arabic-to-English idiom retrieval. Experimental results show that transformer-based classifiers and hybrid retrieval models combining dense embeddings, lexical matching, and cross-encoder reranking achieve strong performance, highlighting the importance of contextual and multilingual representation learning for idiomatic language understanding.

Overall, IdiomX provides a scalable and extensible benchmark for studying figurative language, with a modular pipeline that can be adapted to additional languages and tasks.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. Introduction

Idiomatic expressions represent a fundamental challenge in natural language understanding due to their non-compositional semantics, where the meaning of an expression cannot be inferred directly from its constituent words. For example, expressions such as *kick the bucket* or *spill the beans* require contextual and cultural interpretation rather than literal analysis. This characteristic makes idioms particularly difficult for computational models in tasks such as semantic interpretation, machine translation, information retrieval, and contextual reasoning.

Despite significant advances in large language models (LLMs) and contextual embeddings [6], idiomatic language remains a persistent source of error. While modern models capture general linguistic patterns effectively, they often struggle to distinguish between literal and figurative usage, especially in subtle or ambiguous contexts. A key limitation lies in the availability of suitable datasets. Existing resources are typically limited in scale, lack contextual diversity, or focus on specific tasks and monolingual settings. Datasets such as VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4] have provided valuable benchmarks, but they do not fully address the need for large-scale, context-rich, and multilingual datasets that support both semantic understanding and cross-lingual evaluation.

To address these limitations, we introduce **IdiomX**, a large-scale multilingual benchmark for idiom understanding. IdiomX focuses on English idioms while providing aligned semantic representations in Arabic and French, enabling both monolingual and cross-lingual evaluation. The dataset contains over 190K contextualized examples covering more than 12K idioms, with idiomatic and literal usage labels, multilingual meanings, and rich linguistic metadata. Each idiom is represented across multiple diverse contexts, improving both coverage and evaluation robustness.

A central contribution of IdiomX is its structured and reproducible construction pipeline. The dataset is built through a modular multi-stage process that combines lexical resource extraction, large-scale cleaning and normalization, controlled LLM-based enrichment, and structured validation. The pipeline further incorporates semantic consistency checks, deduplication, and leakage-safe splitting to ensure high data quality and reliable evaluation. In addition, IdiomX integrates multiple data sources, including curated lexical resources,

modern idiomatic usage, and synthetic generation, enabling broad coverage of both traditional and contemporary expressions.

Building on this dataset, we define a unified benchmark consisting of three tasks: (1) idiom detection, (2) context-to-idiom retrieval, and (3) Arabic-to-English idiom retrieval. These tasks collectively evaluate different aspects of figurative language understanding, ranging from classification to semantic retrieval in both monolingual and multilingual settings. Experimental results demonstrate that contextual transformer models achieve strong performance in idiom detection, while hybrid retrieval approaches that combine dense embeddings, lexical matching, and reranking provide substantial improvements for idiom retrieval.

Overall, IdiomX provides a comprehensive and extensible framework for studying idiomatic language, bridging the gap between dataset construction and task-oriented evaluation, and enabling more robust research in figurative and cross-lingual language understanding.

II. Contributions

The main contributions of this work are as follows:

- *IdiomX Dataset* - We introduce IdiomX, a large-scale multilingual idiom benchmark containing over 190K contextualized examples covering more than 12K English idioms. The dataset provides aligned semantic representations in English, Arabic, and French, along with idiomatic and literal usage labels and rich linguistic metadata.
- *Reproducible Multi-Stage Pipeline* - We propose a structured and scalable dataset construction pipeline that integrates curated lexical resources, large-scale cleaning and normalization, controlled LLM-based enrichment, and systematic validation. The pipeline includes semantic consistency checks, deduplication, and leakage-safe data splitting to ensure high data quality and reliable evaluation.
- *Rich Multilingual Semantic Annotations* - We provide high-quality annotations that go beyond surface forms, including contextual meanings, paraphrases, explanations, and cross-lingual semantic alignments. This enables both fine-grained semantic understanding and robust cross-lingual modeling.
- *Unified Multi-Task Benchmark* - We define a comprehensive benchmark consisting of three tasks: (1) idiom detection, (2) context-to-idiom retrieval, and (3) Arabic-to-English idiom retrieval. These tasks collectively evaluate classification, semantic matching, and cross-lingual retrieval capabilities within a unified framework.
- *Strong Baselines and Insights* - We establish competitive baselines across all tasks and provide detailed empirical analyses. Our results demonstrate that contextual transformer models significantly improve idiom detection, while hybrid retrieval approaches combining dense embeddings, lexical signals, and reranking substantially enhance retrieval performance.

III. Related Work

Research on idiomatic language understanding has evolved along two closely related directions: (1) the development of annotated idiom datasets, and (2) modeling approaches for detecting and interpreting idiomatic expressions. These directions are inherently connected, as dataset quality, scale, and diversity directly influence model performance and evaluation reliability.

A. *Idiom Datasets*

Early work on idiom understanding relied on relatively small-scale annotated datasets. Resources such as the VNC-Tokens dataset [1] and the SemEval-2013 Task 5b dataset [2] established foundational benchmarks for distinguishing between idiomatic and literal usage. However, these datasets are limited in size and contextual diversity, which restricts their applicability to modern data-driven approaches.

Subsequent efforts have focused on expanding both coverage and dataset scale. The PIE Corpus [3] introduced a larger collection of idiomatic expressions with contextual annotations, while MAGPIE [4] further improved scale and diversity. Despite these advancements, most existing datasets remain predominantly monolingual and are typically designed for a single task, most commonly idiom detection.

In addition, many datasets focus on specific idiom types, such as verb–noun constructions (e.g., **take place**), fixed expressions (e.g., **by and large**), or semi-compositional phrases (e.g., **spill the beans**). This specialization limits their ability to generalize across the broader spectrum of figurative language.

More recent work has explored multilingual and cross-lingual idiom resources [5, 6]. While these approaches aim to capture semantic equivalence across languages, they often rely on direct translation and lack large-scale contextual examples, balanced idiomatic versus literal supervision, and explicit modeling of ambiguity.

Overall, there remains a clear gap in the availability of large-scale, context-rich, and multilingual idiom datasets that support multiple tasks within a unified evaluation framework. This limitation motivates the development of IdiomX.

B. *Models for Idiom Understanding*

A wide range of computational approaches have been proposed for idiom understanding. Early methods primarily relied on surface-level lexical features and statistical models, such as n-gram representations combined with traditional classifiers. While computationally efficient, these approaches are inherently limited in their ability to capture contextual and non-compositional meaning.

Recent advances have shifted toward neural architectures, particularly transformer-based models such as BERT [13]. These models leverage self-attention mechanisms to capture contextual dependencies between words, enabling more effective disambiguation between literal and idiomatic usage. As a result, transformer-based approaches have become the dominant paradigm for idiom detection and semantic interpretation.

In multilingual settings, cross-lingual embedding models and multilingual transformers have further improved idiom understanding across languages. Methods based on Sentence-BERT [17] and multilingual representation learning frameworks [6] have demonstrated strong performance in semantic similarity and retrieval tasks, enabling alignment between expressions with equivalent meanings across different languages.

Despite these advances, model performance remains highly dependent on the availability of large-scale, high-quality datasets. In the absence of sufficient contextual diversity, balanced supervision, and explicit modeling of ambiguity, even advanced neural models struggle to generalize effectively in idiomatic language understanding. This limitation further

highlights the importance of comprehensive benchmarks such as IdiomX.

C. *Limitations of Existing Work*

Despite substantial progress, several key limitations remain in existing idiom research. First, many datasets are limited in scale, restricting the ability of models to learn robust representations. Second, contextual diversity is often insufficient, with examples lacking variation in style, domain, and linguistic complexity. Third, multilingual and cross-lingual alignment remains underdeveloped, with many resources relying on direct translation rather than semantically grounded representations across languages. Fourth, most existing benchmarks focus on a single task-typically idiom detection—without supporting broader evaluation across multiple dimensions of idiom understanding.

These limitations constrain both model development and evaluation, particularly in scenarios that require nuanced contextual reasoning and cross-lingual semantic understanding. As a result, even advanced models may struggle to generalize beyond narrow settings or specific dataset characteristics.

Addressing these challenges requires large-scale, context-rich, and multilingual resources that support unified evaluation across multiple tasks. This need directly motivates the design and development of IdiomX.

D. IdiomX in Context

To address these challenges, we introduce *IdiomX*, a large-scale multilingual idiom benchmark that integrates several key properties within a unified dataset. IdiomX focuses on English idioms while providing aligned semantic representations in Arabic and French, enabling both monolingual and cross-lingual evaluation. The dataset includes both idiomatic and literal usage examples and supports multiple tasks, including idiom detection, context-to-idiom retrieval, and cross-lingual retrieval.

Compared to prior datasets such as PIE [3] and MAGPIE [4], IdiomX combines large-scale contextual coverage with controlled generation, semantic validation, and minimal duplication (reuse factor ≈ 1.04). Furthermore, each idiom is represented across diverse contextual variations, improving both coverage and evaluation robustness.

These properties position IdiomX as a comprehensive and scalable benchmark for idiomatic language understanding, supporting evaluation across multiple tasks and languages while maintaining high data quality and consistency.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	N	ACL Anthology
SemEval-2013 Task 5b [10]	EN	85	~4,350	N	ACL Anthology
IDIX [9]	EN	52	4,022	N	ACL Anthology
MAGPIE [8]	EN	~1,700	56,000	N	ACL Anthology
PIE-English [11]	EN	~1,200	~20,100	N	arXiv
IdiomX [23]	EN+AR	12k	190k	Y	Public Repository [1]

Table 1: Comparison of IdiomX with existing idiom datasets.

As shown in Table 1, IdiomX surpasses existing datasets in both scale and coverage, providing 190k contextualized examples across 12k idioms. It is the only dataset that combines the following properties:

- large-scale contextual diversity
- bilingual semantic alignment
- unified multi-task support

In contrast to prior resources, which are typically smaller, monolingual, or designed for a single task, IdiomX enables comprehensive evaluation of idiomatic language understanding across multiple dimensions.

This design positions IdiomX as a strong benchmark for idiom detection, semantic retrieval, and cross-lingual language understanding.

IV. DATASET CONSTRUCTION

The IdiomX dataset is constructed through a structured multi-stage pipeline that integrates multiple data sources, controlled large language model (LLM) enrichment, and systematic validation.

Unlike conventional single-source dataset construction approaches, IdiomX is built through 3 complementary pipelines, each targeting a different category of idiomatic expressions: (1) curated lexical resources, (2) modern and informal idiomatic usage, and (3) synthetically generated idioms. These pipelines are subsequently merged, deduplicated, and validated to form a unified high-quality dataset.

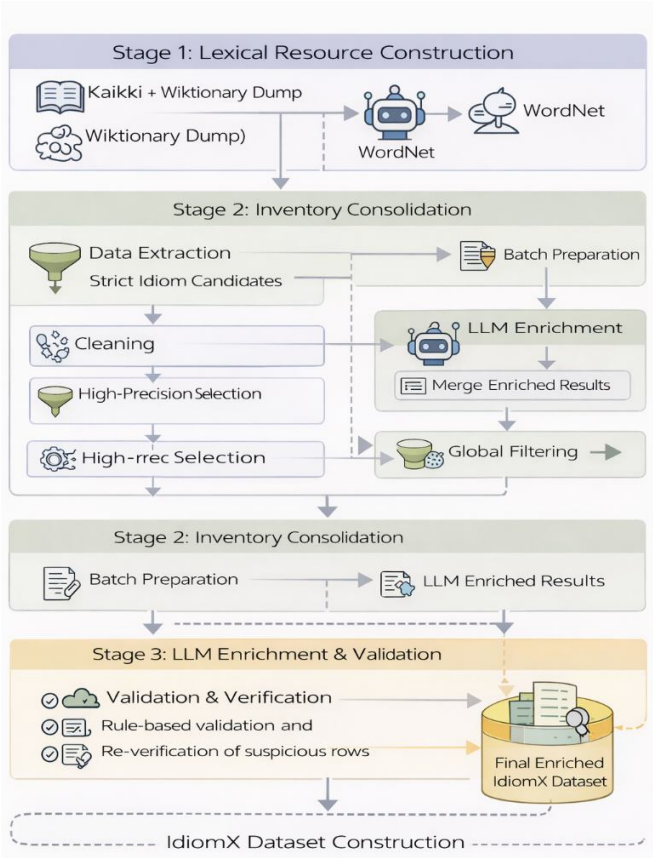


Figure 1 illustrates the overall dataset construction pipeline.

[1] [1] Dataset and code available at:
https://huggingface.co/datasets/aymansharara/IdiomX
https://github.com/aymanshar/idiomx-dataset

A. *Multi-Pipeline Data Collection*

To ensure broad coverage across different types of idiomatic expressions, IdiomX is constructed using three complementary data pipelines:

1) Core Idioms Pipeline

This pipeline aggregates idioms from curated linguistic resources and existing benchmark datasets, including VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4]. These sources provide high-quality, well-defined idiomatic expressions and serve as the foundation of the dataset.

2) Modern Idioms and Slang Pipeline

To capture contemporary and informal language usage, we incorporate idiomatic expressions from modern sources, including slang and emerging expressions. This pipeline improves dataset relevance for real-world applications and introduces linguistic diversity beyond traditional dictionaries.

3) Synthetic Idiom Generation Pipeline

To further expand coverage, we introduce a synthetic generation pipeline using LLMs to create new idiomatic expressions. These generated idioms are subsequently enriched, validated, and filtered to ensure linguistic plausibility and semantic consistency.

After processing each pipeline independently, all collected idioms are merged into a unified representation. Canonical normalization, deduplication, and consistency checks are applied to remove overlaps and ensure a clean and coherent dataset.

B. *LLM-based Dataset Enrichment*

To substantially increase contextual diversity and semantic coverage, we introduce a large-scale enrichment stage using the OpenAI GPT-4.1-mini model [25], selected for its balance between generation quality and computational efficiency.

The enrichment process is implemented as a batch-based asynchronous pipeline, enabling scalable processing of thousands of idioms. Each idiom is transformed into a structured prompt and submitted for generation, allowing high-throughput and reproducible dataset expansion.

For each idiom, the model generates a rich set of structured outputs, including:

- Contextual example sentences across diverse usage scenarios
- Multilingual semantic representations (English, Arabic, and French meanings)
- Natural translations and explanatory descriptions
- Usage labels indicating idiomatic, literal, or borderline interpretations
- Adversarial and challenging examples to increase task difficulty

This stage significantly expands the dataset while preserving **semantic alignment** between idioms, contexts, and meanings, ensuring suitability for both classification and retrieval tasks.

C. *Prompt Design*

To ensure consistency, linguistic quality, and semantic correctness, we employ a structured prompt design strategy. Prompts are carefully engineered to enforce explicit constraints on generation, reducing variability and minimizing noise.

The prompt design enforces the following conditions:

- The idiom must be used naturally within context, avoiding artificial or template-like constructions
- Generated sentences must be grammatically correct and contextually meaningful
- Multilingual meanings must maintain cross-lingual semantic equivalence
- Usage labels must accurately reflect whether the sentence is idiomatic, literal, or ambiguous
- Explanations must be consistent with the intended interpretation of the idiom

In addition, adversarial examples are explicitly incorporated to simulate challenging edge cases, such as semantically ambiguous contexts or misleading lexical cues. This improves the robustness of downstream models, particularly in detection and retrieval tasks.

D. *Automated Validation and Correction*

To ensure data quality at scale, all generated instances undergo a second-stage validation process using the same LLM in a structured verification mode.

Each generated instance is evaluated against strict criteria, including:

- Linguistic correctness
- Semantic consistency between context and meaning
- Alignment between usage label and actual sentence interpretation
- Structural completeness of all required fields

Validation outputs are enforced using a predefined JSON schema [25], ensuring standardized, machine-readable formatting across the dataset.

Each instance is assigned one of the following validation statuses:

- **verified** - accepted without modification
- **corrected** - automatically refined to resolve minor inconsistencies
- **invalid** - rejected due to major semantic or structural errors

This validation stage acts as a quality control layer, reducing noise and ensuring that only high-quality, semantically coherent examples are retained in the final dataset.

The enrichment process follows a schema-guided generation paradigm, where all outputs are constrained to predefined fields to ensure structural consistency.

E. *Quality Control Strategy*

To ensure high data quality and reliability, IdiomX incorporates a multi-layered quality control framework applied throughout the dataset construction pipeline.

1. **Pre-filtering**

- Removal of non-idiomatic or noisy expressions during initial collection
- Deduplication of identical or near-duplicate examples

2. **LLM-based Generation Constraints**

- Use of structured prompt design with explicit linguistic constraints
- Enforcement of schema-guided outputs to ensure consistency across generated fields

3. **Post-generation Validation**

- Automated correction of structurally or semantically inconsistent entries
- Detection of grammatical errors and misaligned usage labels

4. **Statistical Validation**

- Analysis of label distribution, sentence length, and idiom coverage
- Verification of dataset balance and consistency across splits [26]

This layered approach ensures that quality is maintained both at the instance level and at the dataset level, reducing noise while preserving diversity.

F. Examples of Dataset Quality

Idiomatic example
"He finally kicked the bucket after a long illness."
Meaning: To die.

Literal example
"She kicked the bucket across the yard."

Low-quality example (rejected or corrected)
"He kicked bucket yesterday very sadness."
This example is rejected due to grammatical and semantic errors.

G. Final Dataset Characteristics

The final version of IdiomX contains 190k contextualized examples covering 12k English idioms.

Each idiom is associated with multiple contextual instances, ensuring strong diversity in usage patterns. The dataset maintains a balanced distribution between idiomatic and literal examples and includes aligned multilingual semantic representations in English, Arabic, and French, enhancing its suitability for cross-lingual tasks.

V. DATASET ANALYSIS

In this section, we analyze the IdiomX dataset to understand its linguistic structure, quality, and suitability for idiom understanding tasks. The analysis focuses on the full dataset, while observations are consistent across the balanced and high-quality variants unless otherwise stated.

A. Linguistic Properties

The dataset contains over 2.2 million tokens with substantial lexical diversity, supporting robust language modeling.

- **Compositionality Distribution**

Compositionality describes how much the meaning of an expression can be inferred from its individual words. As shown in Fig. 2, most idioms in the dataset are semi-opaque (around 55%), followed by opaque expressions (about 33%) and a smaller portion of transparent cases (around 12%).

This reflects natural language usage, where idioms are often partially interpretable rather than fully literal or completely opaque. This distribution ensures that models must handle different levels of semantic complexity.

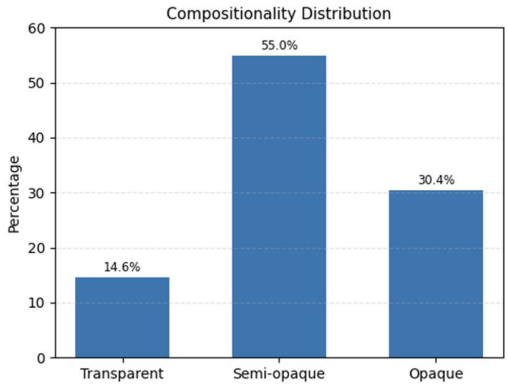


Figure 2: compositionality distribution.

- **Difficulty Distribution**

The learner difficulty distribution, shown in Fig. 3, indicates that most idioms fall into the medium difficulty category (around 78%), with smaller portions of easy and hard examples.

This highlights the inherently challenging nature of idiomatic expressions and provides a realistic benchmark for evaluating model performance.

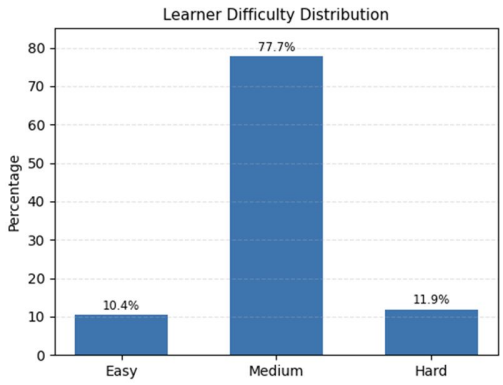


Figure 3: idiom learner difficulty distribution.

B. Data Quality

The dataset exhibits minimal duplication, with a reuse factor close to 1.04, ensuring high diversity and reducing memorization effects.

- **Semantic Quality Distribution**

As shown in Fig. 4, most examples in the dataset are labeled as high quality, with only a small proportion of medium and low-quality samples.

This confirms the effectiveness of the LLM-based generation and validation pipeline in producing reliable and meaningful data. The high-quality variant further enforces strict filtering while preserving the overall dataset characteristics

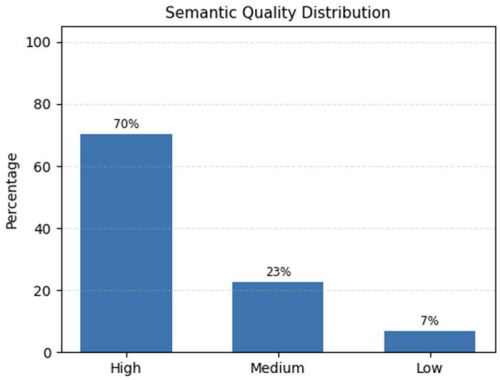


Figure 4: semantic quality distribution.

- **Sentence Length Distribution**

The sentence length distribution, illustrated in Fig. 5, is centered around moderate-length contexts, with an average of approximately 13 words per sentence.

The dataset includes both shorter and longer examples, providing a balanced range of contextual complexity. This ensures that models are exposed to realistic language patterns.

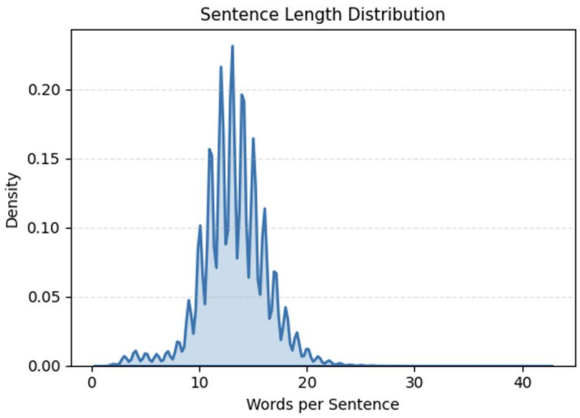


Figure 5: sentence length distribution.

- **Semantic Similarity Distribution**

Figure 6 presents the semantic similarity across different labels. A clear pattern is observed:

- **Literal examples** show the highest similarity
- **Idiomatic examples** show moderate similarity
- **Borderline cases** show lower similarity

This structured gradient introduces controlled difficulty and supports fine-grained evaluation of semantic understanding

Additionally, the high-quality variant consistently improves similarity scores across all categories, confirming that filtering enhances semantic alignment without altering the dataset structure.

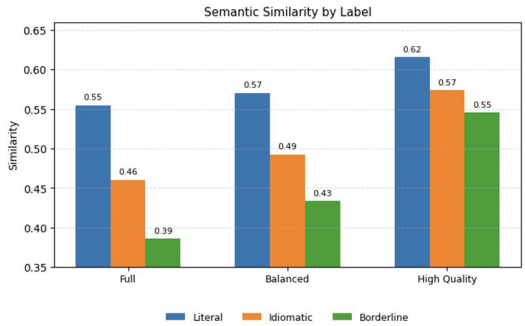


Figure 6: Semantic Similarity by Label.

C. Label and Ambiguity Distribution

- **Idiomatic vs Literal Balance**

The dataset maintains a near-balanced distribution between idiomatic and literal usage (around 46% each), with a small proportion of borderline cases.

This balance is critical for tasks involving semantic disambiguation, where models must distinguish between literal and figurative meaning

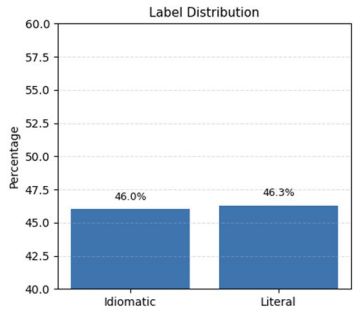


Figure 7: Label Distribution.

- **Ambiguity**

As shown in Fig. 8, the dataset captures a graded view of idiomaticity.

- A large portion of examples are semi-literal possible
- A significant portion are strongly idiomatic
- A smaller portion are fully ambiguous

This reflects realistic language behavior, where meaning often depends on context rather than fixed interpretation.

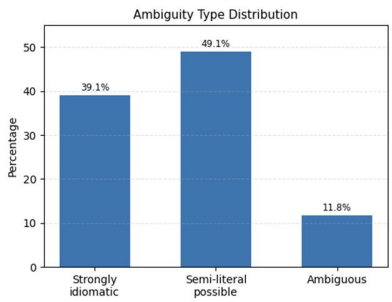


Figure 8: *Ambiguity Type Distribution*

D. ***Final Dataset Release***

IdiomX is released as a single consolidated dataset, designed to maximize coverage while maintaining high linguistic quality and consistency.

The final dataset contains approximately 190k contextualized examples, covering over 12k English idioms,

with a large proportion of unique sentences and minimal duplication. Each idiom is associated with multiple contextual instances, ensuring diverse usage patterns across different linguistic scenarios.

VI. Benchmark Tasks

To comprehensively evaluate idiom understanding, we define three benchmark tasks that capture complementary aspects of figurative language processing, including classification, contextual reasoning, and cross-lingual semantic alignment.

Together, these tasks form a unified evaluation framework that moves beyond isolated settings and enables systematic analysis of idiomatic language understanding in both monolingual and multilingual contexts.

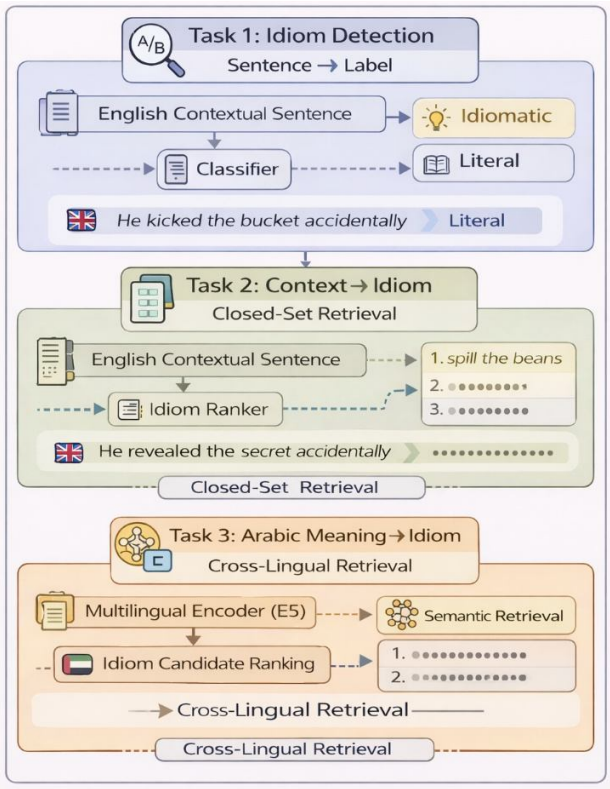


Figure 9: *Idiomx 3 benchmark tasks*

A. Task 1: Idiom Detection

Idiom detection focuses on identifying whether an expression in a sentence is used idiomatically or literally, a well-known challenge in figurative language understanding [1], [2].

We formulate this task as a binary classification problem.

Each instance consists of:

- **Input:** a sentence containing a target expression
- **Output:** a label ∈ {idiomatic, literal}

The task evaluates a model's ability to distinguish figurative meaning from literal interpretation, particularly in cases where the same expression can exhibit dual meanings depending on context (e.g., "kick the bucket").

This task primarily measures contextual disambiguation capability, which is essential for downstream applications such as machine translation, semantic parsing, and dialogue systems.

B. Task 2: Context-to-Idiom Retrieval

Given a contextual sentence, the goal is to retrieve the idiom that best captures the underlying meaning expressed in the context.

We formulate this as a closed-set retrieval problem over a predefined idiom inventory.

Each instance consists of:

- **Input:** a contextual sentence
- **Output:** a ranked list of candidate idioms

The objective is to rank the correct idiom at the top of the candidate list.

This task evaluates deep semantic understanding and expressive alignment, requiring models to map contextual meaning to an appropriate figurative expression rather than relying on surface-level lexical cues.

C. Task 3: Arabic-to-English Idiom Retrieval

This task extends retrieval to a cross-lingual setting, where the input is an Arabic sentence and the goal is to retrieve the corresponding English idiom.

We formulate this as a ranking problem, where candidate idioms are scored based on their semantic similarity to the input.

Each instance consists of:

- **Input:** an Arabic contextual sentence
- **Output:** a ranked list of English idioms

This task evaluates a model's ability to perform cross-lingual semantic alignment, particularly for figurative expressions that often lack direct lexical correspondence [6].

It is especially challenging due to differences in cultural context, expression structure, and semantic abstraction across languages.

D. *Unified Evaluation Perspective*

Together, these tasks form a progressive evaluation framework:

- **Task 1** → contextual classification (idiomatic vs. literal)
- **Task 2** → monolingual semantic retrieval (context → idiom)
- **Task 3** → cross-lingual semantic retrieval (Arabic → English idiom)

This progression enables systematic evaluation of models across increasing levels of difficulty, from surface-level disambiguation to deep semantic reasoning and cross-lingual understanding.

VII. EXPERIMENTS AND RESULTS

This section presents the experimental evaluation of the IdiomX dataset across the defined benchmark tasks. The goal is to assess how well the dataset supports different problem types, including idiom detection, context-to-idiom retrieval, and cross-lingual retrieval.

We evaluate both classification and retrieval performance using standard metrics widely used in natural language processing. The results provide insights into how different modeling approaches handle idiomatic language, which often requires deeper contextual understanding beyond surface-level text.

A. Experimental Setup

The IdiomX dataset is split into training, validation, and test sets using a stratified approach to ensure balanced distribution of idiomatic and literal examples across all splits.

Each idiom appears in both idiomatic and literal contexts, creating a controlled and fair evaluation setting.

For classification tasks, we report:

- Accuracy
- Precision, Recall, and F1-score

For retrieval tasks, we use:

- Top-1 accuracy
- Top-K accuracy

These metrics allow evaluation of both exact predictions and ranking quality.

We evaluate three types of models:

- **TF-IDF + Logistic Regression (lexical baseline)** - captures surface-level patterns without contextual understanding.
- **DistilBERT [12]** - a lightweight transformer model used as a strong baseline.
- **RoBERTa [14]** - a larger transformer model designed to capture deeper contextual relationships.

For cross-lingual tasks, multilingual models such as XLM-R [6] are used, as they are designed to handle multiple languages effectively.

All experiments are conducted using the same preprocessing pipeline and consistent training settings to ensure a fair comparison between models.

We begin with Task 1 (Idiom Detection), which serves as the primary classification benchmark for evaluating contextual understanding

B. *Task 1: Idiom Detection*

1. *Problem Definition*

The goal of this task is to determine whether a sentence contains an idiomatic or a literal usage of an expression. This is challenging because idiomatic meaning cannot be directly inferred from individual words and depends heavily on context [1], [2].

We formulate this as a binary classification problem, where each sentence is labeled as either *idiomatic* or *literal*.

2. *Models*

We evaluate three models with increasing levels of contextual understanding:

• **TF-IDF + Logistic Regression (Baseline)**

A traditional machine learning approach that represents text using word frequency features. This model captures surface-level lexical patterns but does not model context.

• **DistilBERT [12] (Transformer Baseline)**

A compact transformer model that uses self-attention to capture contextual relationships between words. It provides a strong balance between performance and efficiency.

• **RoBERTa [14] (Improved Model)**

A larger and more powerful transformer model trained with optimized pretraining strategies. It is designed to capture deeper contextual and semantic relationships in text.

All models are trained and evaluated under the same conditions to ensure a fair and consistent comparison.

3. *Data Processing*

Input sentences were tokenized using the pretrained tokenizer corresponding to each model. The following preprocessing settings were applied:

- Padding to a fixed maximum length of 128 tokens
- Truncation of longer sequences

The dataset was split using a stratified approach to ensure balanced distribution of idiomatic and literal examples. The Hugging Face *datasets* library was used for efficient preprocessing and data handling.

label	Count	Percentage (%)
literal	90,041	50.01
idiomatic	89,722	49.99

Table 3: Class Distribution of Idiomatic vs Literal Examples.

The dataset is well balanced, ensuring that evaluation is not biased toward a dominant class.

4. *Training Setup*

All models were fine-tuned using the same configuration:

- Batch size: 8
- Learning rate: 2e-5
- Number of epochs: 3
- Evaluation performed at the end of each epoch
- Best model selected based on validation F1-score

This setup ensures stable training and enables a fair comparison between models.

5. *Results*

- *Model Comparison*

Model	Accuracy	F1-score
TF-IDF + Logistic egression	0.865	0.865
DistilBERT	0.919	0.919
RoBERTa	0.926	0.926

Table 2: Model Performance on Idiom Detection Task

Transformer-based models significantly outperform the lexical baseline, demonstrating the importance of contextual understanding for idiom detection.

RoBERTa achieves the best performance, with an accuracy of 93.06% and F1-score of 0.929, confirming the benefit of larger and more expressive models.

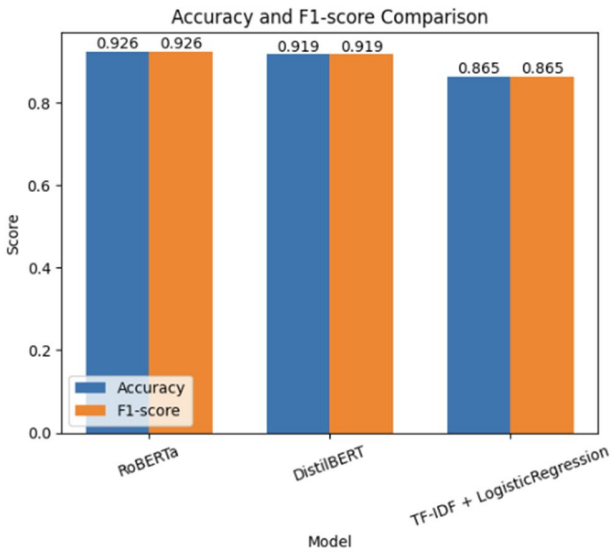


Figure 10: Model Accuracy and F1-score comparison on idiom detection

DistilBERT provides a strong trade-off between performance and efficiency, achieving competitive results despite its smaller size

- *Detailed Performance Analysis*

RoBERTa shows consistent improvement over DistilBERT across all metrics. In particular:

• higher precision → fewer false positives
• balanced recall → stable detection of idiomatic usage
• improved F1-score → better overall classification quality

This indicates that deeper contextual representations are critical for distinguishing subtle differences between literal and idiomatic meaning.

6. *Error Analysis and Key Failure Modes*

To better understand model limitations, we analyze prediction errors using both quantitative and qualitative approaches. In addition to examining misclassified examples, we analyze confidence scores and error distributions to identify systematic patterns:

(1) Limited or Insufficient Context

Short or context-poor sentences make it difficult to determine whether an expression is used idiomatically or literally. Without sufficient surrounding information, the model may rely on surface patterns rather than meaning.

(2) Strong Idiom Bias

Highly frequent idioms (e.g., “kick the bucket”) are sometimes predicted as idiomatic even when used literally. This indicates that the model partially memorizes common idioms and overgeneralizes them across contexts.

This behavior is also reflected in the confusion matrix, where false positives are slightly more frequent than false negatives

(3) Context-Dependent Ambiguity

Some expressions can be interpreted as either literal or idiomatic depending on subtle contextual cues. These borderline cases are particularly challenging, as they require fine-grained semantic understanding beyond explicit wording.

(4) Indirect or Implied Meaning

In certain cases, idiomatic meaning is not explicitly stated but implied. The model may fail to capture such implicit meanings, leading to false negatives.

(5) Overconfidence in Incorrect Predictions

Analysis of prediction confidence reveals that the model often assigns high confidence scores even to incorrect predictions.

The mean confidence for incorrect predictions is approximately **0.908**, while correct predictions reach approximately **0.988**, indicating strong confidence separation but also highlighting cases of overconfidence.

```
Correct mean: 0.9874
Incorrect mean: 0.9042
```

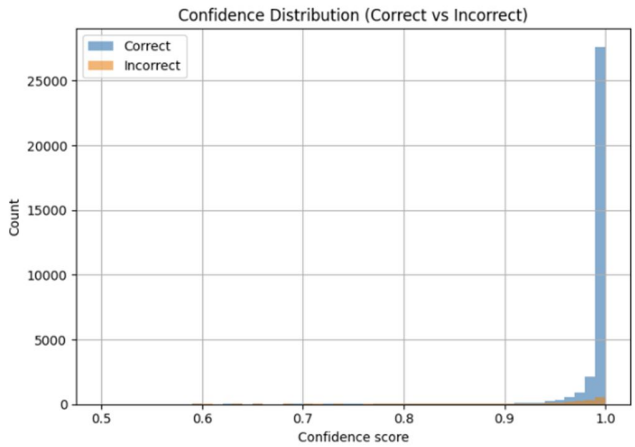
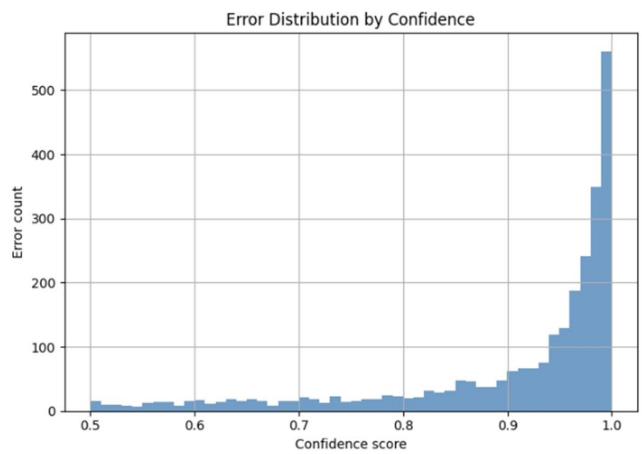


Figure 11: Confidence distribution for correct vs incorrect predictions.

As shown in Figure 11, most correct predictions are concentrated near high confidence values, while incorrect predictions, although fewer, still appear at high confidence levels.

(6) Error Distribution by Confidence

To further analyze model reliability, we examine how errors are distributed across confidence scores.




```
Mean error confidence: 0.9042
```

Figure 12: Error distribution by confidence.

Most errors occur at high confidence levels, indicating that the model is often confidently wrong in difficult or ambiguous cases. This suggests that confidence alone is not always a reliable indicator of correctness.

(7) Hard Negative Analysis

We analyze the most confident incorrect predictions (hard negatives) to better understand challenging cases.

	example	y_true	y_pred	confidence
14488	At first blush, the committee's reputation gai...	0	1	0.999653
22463	Oh sure, that book really blew her back out wi...	0	1	0.999636
148856	she opened the world like an oyster	0	1	0.999608
132339	My boss just slopped out a whole month's work ...	0	1	0.999604
124023	The CEO ran off with the company's confidentia...	0	1	0.999599
170083	Miner: If I strike gold here, it'll change my ...	0	1	0.999594
81103	Just saw my neighbors having a wild party—wish...	0	1	0.999588
48052	OMG! That concert was wow-worthy, add an excla...	0	1	0.999564
118990	Oh sure, she's a real race queen—she won the n...	0	1	0.999542
136430	The university's event stood us up by unexpect...	0	1	0.999542

Figure 13: Examples of high-confidence errors.

Many hard negatives involve sentences where:
• both literal and idiomatic interpretations are plausible
• subtle contextual cues determine the correct meaning
• the model fails to fully capture semantic nuance

In several cases, prediction confidence exceeds **0.99**, indicating strong overconfidence in ambiguous contexts.

Overall, these findings show that while transformer models capture contextual information effectively, they still struggle with ambiguity, implicit meaning, and fine-grained semantic distinctions.

These observations are consistent with prior work on figurative language understanding [3], which highlights the difficulty of modeling non-compositional meaning.

7. Qualitative Examples

To further illustrate model behavior, we present representative examples covering both correct predictions and common failure cases:

Example 1 (Correct – Idiomatic)
"He finally kicked the bucket after a long illness."
Prediction: Idiomatic

Analysis:

The model correctly identifies the figurative meaning based on strong contextual cues indicating death. This demonstrates effective contextual understanding for well-formed idiomatic usage.

Example 2 (Correct – Literal)
"She kicked the bucket across the yard."
Prediction: Literal

Analysis:

The sentence clearly describes a physical action, and the model correctly distinguishes it from idiomatic usage. This indicates that the model can recognize explicit literal contexts.

Example 3 (Error – False Positive)
"He kicked the bucket lightly to move it aside."
Prediction: Idiomatic
Actual: Literal

Analysis:

The model incorrectly predicts idiomatic usage due to the presence of a highly frequent idiom. This reflects a bias toward memorized patterns, where the model over-relies on the expression itself rather than fully interpreting the surrounding context.

Example 4 (Error – False Negative)
"After years of stress, he finally let everything go."
Prediction: Literal
Actual: Idiomatic

Analysis:

The idiomatic meaning is implied rather than explicitly stated, making it more difficult to detect. This highlights a limitation in capturing subtle or indirect figurative expressions.

These examples confirm that most errors arise from contextual ambiguity and implicit meaning rather than random misclassification.

In particular:

- false positives are often driven by idiom memorization bias
- false negatives are linked to implicit or weak contextual signals

These observations are consistent with the error patterns discussed in the previous section and further emphasize the challenge of modeling non-compositional language.

8. Confusion Matrix Analysis

The confusion matrix provides a detailed view of model performance across idiomatic and literal classes.

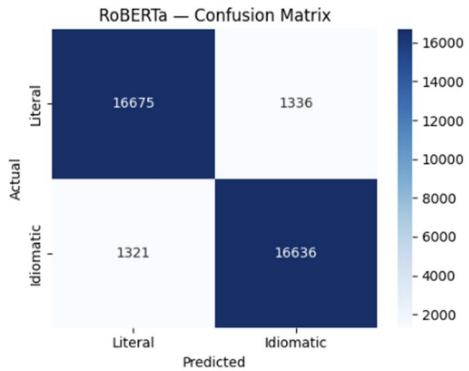


Figure 14: Confusion matrix for RoBERTa on the test set.

As shown in Figure 14, the model achieves strong performance for both classes, with high true positive rates for both idiomatic and literal examples.

However, a slightly higher number of false positives can be observed (literal → idiomatic). This indicates a mild tendency of the model to classify expressions as idiomatic, even when they are used literally.

This behavior is consistent with the previously identified idiom frequency bias, where well-known expressions are more likely to be interpreted as idiomatic.

Overall, the confusion matrix confirms balanced performance, with a small but consistent bias toward idiomatic predictions.

9. ROC Curve Analysis

To further evaluate classification performance across different decision thresholds, we analyze the Receiver Operating Characteristic (ROC) curve.

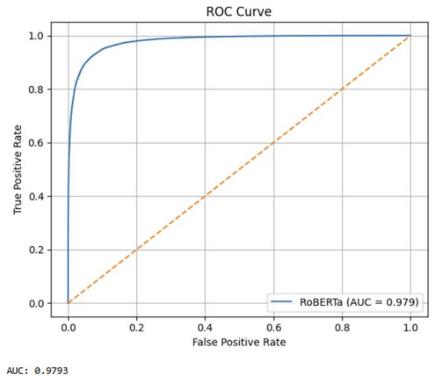


Figure 15: ROC curve for RoBERTa (AUC = 0.979).

The ROC curve shows a strong separation between classes, with an Area Under the Curve (AUC) of **0.979**, indicating excellent discriminative ability.

The curve remains close to the top-left corner, demonstrating that the model achieves a high true positive rate while maintaining a low false positive rate across thresholds.

This result confirms that the model is robust and reliable in distinguishing between idiomatic and literal usage, even when decision thresholds vary.

10. Ablation Study

To understand the impact of model capacity, we compare the performance of DistilBERT and RoBERTa.

RoBERTa achieves higher accuracy and F1-score, indicating stronger contextual understanding and better ability to capture subtle semantic differences.

In contrast, DistilBERT achieves slightly higher recall, suggesting that it is more permissive and tends to classify more examples as idiomatic, leading to increased false positives.

This trade-off reflects a common pattern in transformer models:

• smaller models generalize more aggressively
• larger models produce more precise and balanced predictions

These results confirm that increased model capacity improves idiom detection performance, particularly for capturing nuanced contextual cues.

11. Conclusion

The results demonstrate that transformer-based models are highly effective for idiom detection, where contextual understanding is essential. While DistilBERT provides a strong and efficient baseline, RoBERTa consistently achieves the best performance across all key metrics, including accuracy and F1-score.

These findings show that idiomatic language cannot be reliably modeled using surface-level patterns alone, and instead requires deeper contextual and semantic representations. Larger transformer architectures are better suited to capture these nuances, leading to more accurate and balanced predictions.

In addition, the analysis of confidence scores and hard negative examples highlights that ambiguity and implicit meaning remain key challenges, even for high-performing models

Overall, the results confirm that IdiomX provides a robust and reliable benchmark for idiom detection. The dataset enables comprehensive evaluation of contextual language understanding and supports the development of more advanced models for figurative language processing.

C. Task 2: Context-to-Idiom Retrieval

Task 2 evaluates whether a model can recover the correct canonical idiom from an English contextual sentence that uses the idiom idiomatically. Unlike Task 1, which is a binary classification problem, Task 2 is formulated as a closed-set retrieval problem over a fixed idiom inventory.

Formally, given a contextual query x, the goal is to retrieve the correct idiom y* from a fixed idiom inventory I:

$$y^* = \arg\max_{y \in I} score(x, y)$$

where $\text{score}(x, y)$ denotes the retrieval score assigned to idiom y given context x.

This formulation requires both semantic understanding of the contextual sentence and precise matching to a canonical idiom form. Because the candidate inventory is fixed, the task provides a controlled and reproducible benchmark for evaluating retrieval-based idiom understanding.

1. Benchmark Construction and Split Strategy

The Task 2 benchmark is constructed from the IdiomX high-quality subset. To ensure a reliable retrieval setting, we retain only examples that satisfy the following conditions:

(1) the usage label is idiomatic,
(2) validation_status = valid, and
(3) semantic_quality = high.

After filtering, the final benchmark contains 54,469 examples associated with an idiom bank of 13,803 canonical idioms.

To prevent sentence leakage between training and evaluation, we apply a query-level split. Under this strategy, the same contextual sentence is never shared across splits. However, the same idiom may appear in both training and test sets through different contextual realizations, which is appropriate for the closed-set retrieval setting because the target inventory is fixed at evaluation time.

The final split contains 43,601 training examples and 10,868 test examples.

Setting	Value
Total examples	54,469
Idiom bank size	13,803
Train examples	43,601
Test examples	10,868
Split strategy	Query-level
Sentence leakage across splits	None

Table 4: Task 2 benchmark statistics and split configuration.

2. Retrieval Architecture

We evaluate three retrieval settings of increasing complexity.

- **MiniLM dense retrieval**.

In the first setting, both contextual sentences and candidate idioms are encoded into a shared embedding space using MiniLM sentence embeddings [28]. Retrieval is performed by ranking idioms according to semantic similarity between the input context and idiom representations.

- **Hybrid retrieval.**

In the second setting, dense retrieval is combined with BM25 [27] sparse retrieval. The dense component captures semantic similarity beyond surface wording, while the sparse component captures lexical overlap. Their combination improves candidate recall by leveraging both semantic and lexical signals.

- **Hybrid retrieval with reranking.**

In the third setting, a cross-encoder reranker [29] is applied to the top candidates returned by the hybrid retriever. The reranker jointly scores each context–idiom pair, allowing more precise discrimination among semantically similar candidates.

- **Hybrid Retrieval with Fine-Tuned Reranker.**

o further improve ranking quality, we fine-tune the cross-encoder reranker on task-specific query–idiom pairs. The model is initialized from a pre-trained Sentence-Transformers cross-encoder (ms-marco-MiniLM-L-6-v2) and trained using a pairwise ranking objective, where the ground-truth idiom is treated as a positive example and retrieved candidates serve as hard negatives.

Fine-tuning enables the model to capture subtle semantic distinctions between closely related idioms and significantly improves ranking precision.

This multi-stage design follows a standard retrieval pipeline in which early stages focus on recall, while later stages improve ranking accuracy [29].

3. Evaluation Metrics

Performance is evaluated using standard ranking metrics:

- **Top-1 Accuracy**: correct idiom ranked first
- **Top-3 / Top-5 Accuracy**: correct idiom appears in top-k candidates
- **Mean Reciprocal Rank (MRR)**: average reciprocal rank of the first correct prediction.

Top-k accuracy captures retrieval success at different cutoffs, while MRR provides a more fine-grained measure of ranking quality.

4. Quantitative Results

able 5 reports the results for Task 2 under the closed-set retrieval setting.

Model	Top-1	Top-3	Top-5	MRR
MiniLM	0.640	0.748	0.787	0.707
Hybrid (MiniLM + BM25)	0.761	0.857	0.882	0.817
Hybrid + reranker	0.838	0.901	0.906	0.869
Hybrid + Fine-Tuned Reranker	0.885	0.909	0.909	0.897

Table 5: Main Task 2 retrieval results on the closed-set benchmark.

he results show a clear and consistent improvement across all evaluation metrics as the retrieval pipeline becomes more selective and expressive.

MiniLM [28] achieves a Top-1 accuracy of 0.640 and an MRR of 0.707, indicating that dense retrieval alone provides a reasonable semantic baseline. However, it struggles with fine-grained ranking among semantically similar idioms, where subtle contextual distinctions are required.

When BM25 is combined with MiniLM [28], performance increases substantially to 0.761 Top-1 and 0.817 MRR, demonstrating that lexical matching and semantic similarity provide complementary signals. This confirms prior findings in retrieval literature that hybrid approaches outperform purely dense models [17], [27].

Adding a cross-encoder reranker further improves performance to 0.838 Top-1 and 0.869 MRR, showing that pairwise scoring is effective for refining candidate ordering. The reranker is particularly beneficial in resolving ambiguity among the top retrieved candidates, where coarse retrieval signals are insufficient

The best performance is achieved with the fine-tuned reranker, reaching 0.885 Top-1, 0.909 Top-3, 0.909 Top-5, and 0.897 MRR. This demonstrates that task-specific fine-tuning enables the model to capture subtle semantic distinctions between closely related idioms, leading to significant improvements in ranking precision

Figure 16 visualizes the overall comparison across Top-1, Top-3, Top-5, and MRR.

Figure 17 further illustrates the Top-K retrieval curves under the closed-set setting.

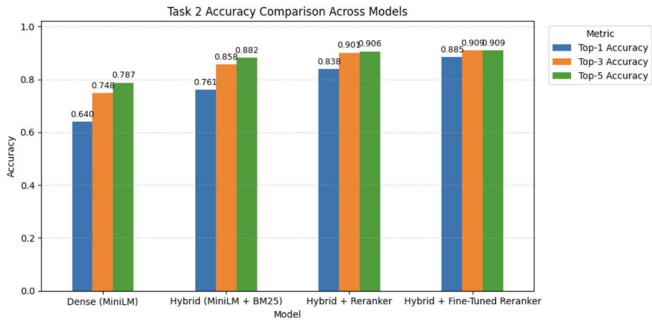


Figure 16: Performance comparison of Task 2 retrieval models across Top-1, Top-3, Top-5, and MRR.

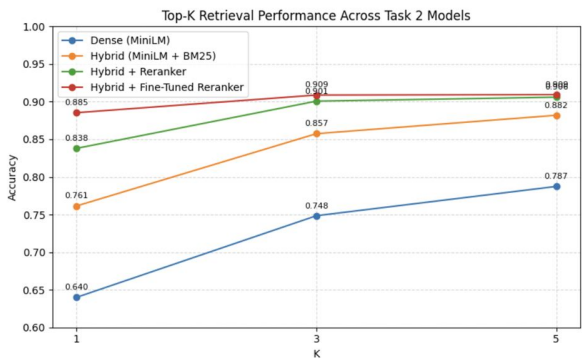


Figure 17: Top-*K* retrieval curves for Task 2 under the closed-set retrieval setting.

Both figures consistently show that each additional retrieval stage improves performance, with the largest gains observed at Top-1, which is the most critical metric for exact idiom recovery.

To quantify improvements over the dense baseline, Table 6 reports the absolute gains relative to MiniLM.

Metric	Hybrid Gain	Hybrid + Reranker	Fine-Tuned Reranker
Top-1 Accuracy	+0.121	+0.198	+0.245
Top-3 Accuracy	+0.109	+0.153	+0.161
Top-5 Accuracy	+0.095	+0.119	+0.122
MRR	+0.110	+0.162	+0.190

Table 6: Absolute improvement over the MiniLM dense retrieval baseline.

These results highlight three key findings:

1. **Hybrid retrieval significantly improves recall**, confirming the importance of combining lexical and semantic signals.
2. **Reranking improves precision**, especially among top candidates.
3. **Fine-tuning delivers the largest gains**, enabling the model to better distinguish between semantically similar idioms.

Overall, the results demonstrate that context-to-idiom retrieval is not solved by semantic embeddings alone, but instead requires a combination of semantic retrieval, lexical matching, and task-specific reranking [17], [27], [29].

5. Error Analysis and Failure Modes

The best-performing model (Hybrid + Fine-Tuned Reranker) achieves strong performance, with 0.885 Top-1 accuracy and 0.897 MRR. However, a detailed analysis of the test set reveals 1,246 incorrect predictions out of 10,868 samples, as summarized in Figure 18. This corresponds to approximately 11.5% residual error, indicating that the task remains challenging despite high overall accuracy.

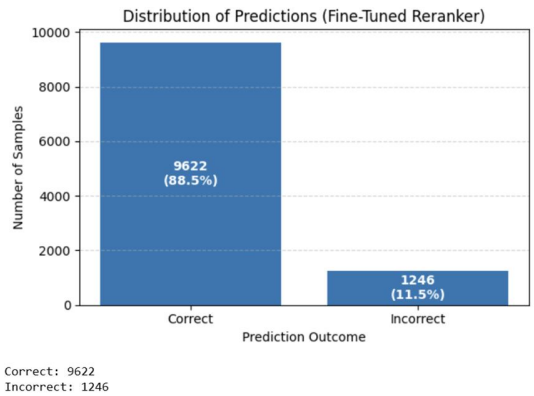


Figure 18: Distribution of correct and incorrect predictions for Task 2 on the test set.

A qualitative inspection of the errors reveals several consistent patterns.

First, the model often confuses idioms that are lexically or semantically similar, selecting plausible but incorrect expressions. This is particularly evident in near-miss cases where the correct idiom appears within the Top-3 predictions but is not ranked first.

Second, there is a noticeable frequency bias, where common idioms are preferred over less frequent but correct ones, a known effect in retrieval systems [27]. This suggests that model scoring is partially influenced by distributional frequency.

Third, the model sometimes relies on surface-level keyword matching, especially when lexical overlap between the context and candidate idioms is strong but semantically misleading. This results in incorrect predictions despite apparent lexical alignment.

Fourth, performance degrades on short or ambiguous contexts, where insufficient contextual information makes it difficult to disambiguate between competing idioms.

Finally, errors are more frequent for rare (long-tail) idioms, indicating that even with fine-tuning, the model struggles with low-frequency expressions that have limited representation during training.

Overall, these errors suggest that while the retrieval pipeline effectively combines semantic and lexical signals, idiom understanding still requires fine-grained contextual reasoning, improved robustness to frequency bias, and better handling of long-tail expressions [17], [29].

6. Qualitative Analysis

To better understand model behavior, we present qualitative examples from the test set illustrating both successful retrievals and typical failure cases. Table 7 shows representative examples produced by the best-performing model (Hybrid + Fine-Tuned Reranker).

The examples highlight that the model is generally able to recover the correct idiom when the contextual signal is strong and unambiguous. In such cases, the correct idiom is ranked at the top position, demonstrating effective semantic matching between the input context and the idiom meaning.

However, failure cases reveal consistent challenges. In particular, the model tends to:

- select idioms that are lexically similar but semantically incorrect,
- prefer more frequent expressions that partially match the context,
- struggle with ambiguous or weak contextual cues.

These errors are consistent with the patterns identified in Section 5, where lexical overlap and frequency bias can override deeper semantic understanding

Overall, the qualitative results show that the retrieval pipeline performs robustly in clear contexts but struggles in cases requiring fine-grained semantic discrimination or involving less frequent idioms.

Table 7 presents representative examples from the best-performing model. The first two cases illustrate typical failure modes, while the final example shows a correct prediction under clear contextual conditions.

Query (shortened)	True Idiom	Top-1	Top-3 (shortened)	✓
"…eat them for breakfast."	eat someone for breakfast	don't worry	don't worry; eat someone…; take one…	✗
"Bone apple tea… bon appétit"	bon appétit	throw a bone to	throw a bone…; wise apple; contrary…	✗
"…finally broke the ice…"	break the ice	break the ice	break the ice; ease tension; start talking	✓

Table 7: Representative qualitative examples from Task 2

7. Task 2 Summary

In this task, we introduced a closed-set context-to-idiom retrieval benchmark built on the IdiomX high-quality dataset. The task requires mapping an idiomatic contextual sentence to its corresponding canonical idiom from a fixed inventory, combining semantic understanding with precise lexical grounding.

The results show that retrieval performance improves consistently across the proposed pipeline. Dense retrieval using MiniLM [28] provides a strong semantic baseline, while hybrid retrieval significantly improves performance by incorporating complementary lexical matching signals [17], [27]. The addition of a cross-encoder reranker further enhances ranking quality by refining candidate ordering.

The best results are achieved with the fine-tuned reranker, which reaches 0.885 Top-1 accuracy and 0.897 MRR. This demonstrates that task-specific fine-tuning enables the model to capture fine-grained semantic distinctions between closely related idioms, leading to substantial improvements in ranking precision.

Overall, Task 2 highlights that idiom understanding can be effectively modeled as a retrieval problem, but achieving high performance requires a combination of semantic representations, lexical matching, and task-specific reranking

8. Discussion

The results of Task 2 provide several important insights into idiom understanding when formulated as a retrieval problem..

First, while dense semantic embeddings are effective at capturing contextual meaning, they are not sufficient on their own for precise idiom recovery. The consistent improvement observed with hybrid retrieval confirms that lexical signals remain essential, even in semantically driven tasks [17], [27]. This is particularly relevant for idioms, where surface forms often encode distinctive cues that complement contextual semantics.

Second, the reranking stage plays a critical role in refining predictions. By jointly modeling interactions between the input context and candidate idioms, the cross-encoder significantly improves ranking precision and resolves ambiguities among top candidates [29]. The strong gains achieved by the fine-tuned reranker further demonstrate the importance of task-specific optimization for capturing subtle semantic distinctions between closely related expressions.

Third, the error analysis indicates that the remaining challenges are primarily driven by semantic ambiguity, frequency bias, and limited contextual information. These findings suggest that future improvements should focus on more robust contextual representations, improved handling of long-tail idioms, and training strategies that mitigate bias toward frequent expressions.

Finally, the proposed benchmark provides a controlled and reproducible evaluation framework, enabling consistent comparison across retrieval architectures. As such, it offers a solid foundation for future research on idiom understanding, retrieval-based semantic modeling, and multilingual extensions.

D. Task 3: Arabic Context-to-Idiom Retrieval

This task evaluates whether a model can recover the correct canonical English idiom from an Arabic contextual sentence. Similar to Task 2, this is formulated as a closed-set retrieval problem over a fixed idiom inventory. However, unlike Task 2, this setting introduces a cross-lingual challenge, where the input is in Arabic and the target idiom is in English.

Formally, given an Arabic query x, the goal is to retrieve the correct idiom $y^*$from a fixed idiom inventory I:

$$y^* = \arg\max_{i \in I} sim(x, i)$$

where $\mathrm{sim}(x, i)$is the similarity score between the Arabic context and the candidate idiom.

This formulation requires both semantic understanding of the Arabic context and accurate alignment to a canonical English idiom. Compared to Task 2, the task is more challenging due to cross-lingual variation and differences in idiomatic expression across languages.

As in Task 2, we adopt a retrieval-based formulation because the candidate idiom inventory is fixed, enabling a controlled and reproducible evaluation setting.

1. Formulation

Each Arabic query $x$and candidate idiom $i$are mapped into vector representations using a shared encoder:

$$h_x = Encoder(x), \qquad h_i = Encoder(i)$$

Similarity is computed using cosine similarity [5]:

$$sim(x, i) = \frac{h_x \,.\, h_i}{\|h_x\| . \|h_i\|}$$

Prediction is defined as:

$$\hat{y} = \arg\max_{i} sim(x, i)$$

where cosine similarity [5] is used to rank candidate idioms in the shared embedding space.

2. Experimental Setup

To construct the Task 3 benchmark, we initially explored multiple approaches, including generation-based models (mT5 [19]), hybrid generation–retrieval variants, and Arabic semantic bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple fields and applied a strict split strategy to prevent overlapping at the idiom level.

While this setup was methodologically rigorous, it resulted in near-zero retrieval accuracy, indicating that the task formulation was overly constrained and not well aligned with a retrieval-based setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem. The final benchmark retains only idiomatic examples and uses idiom_in_example_arabic as the Arabic query and idiom_canonical as the English retrieval target.

Instead of splitting by idiom identity, we apply an example-level split, allowing the same idiom to appear in both training and test sets through different contextual realizations. This setup evaluates whether the model can generalize to unseen Arabic contexts while preserving a realistic and fixed candidate inventory.

Under this revised formulation, retrieval performance improves substantially. The fine-tuned multilingual E5 model achieves strong results (Top-1 = 64.86%, Top-5 = 81.46%), demonstrating effective cross-lingual semantic alignment.

Overall, the results show that stronger pretrained multilingual embeddings alone provide only modest gains, while task-specific contrastive fine-tuning is the primary driver of performance improvement. Additional gains are obtained through richer supervision, such as multiple positive examples per query and extended training.

The final best-performing system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. *Candidate Representation*

In Task 3, the candidate idiom inventory is represented using multiple Arabic textual views to improve semantic coverage. For each idiom, we include:

- example sentences (idiomatic usage)
- example meanings
- canonical meanings

Each representation is encoded into a shared multilingual embedding space, producing a dense representation matrix over the idiom inventory.

This multi-view representation allows the model to match Arabic queries against diverse contextual signals, improving robustness in cross-lingual retrieval.

4. *Models Evaluated*

We evaluate a set of multilingual dense retrieval models of increasing capability:

- Multilingual MiniLM [28] (baseline)
- Multilingual E5 (zero-shot)
- Multilingual E5 (fine-tuned)

MiniLM [28] serves as a lightweight semantic baseline. The E5 model introduces improved multilingual representation quality, while the fine-tuned variant is trained using contrastive learning on Arabic–idiom pairs to better align cross-lingual semantics.

While all models are evaluated, detailed analysis focuses on the best-performing configuration.

5. *Quantitative Results*

Performance is evaluated using standard ranking metrics, including Top-1 and Top-5 accuracy. Top-K accuracy measures whether the correct idiom appears within the top Kretrieved candidates.

$$Top - K = \frac{1}{N}\sum 1\{y \in TopK(x)\}$$

Table 8 reports the retrieval performance of the evaluated models.

Model	Embedding Dim	Top-1	Top-5
MiniLM-L12-v2 [28]	384	0.159	0.254
E5-base (zero-shot)	768	0.191	0.310
E5-base (finetuned)	768	0.578	0.761

Table 8: Retrieval performance on Task 3.

The results show a clear and consistent improvement across all models. MiniLM [28] provides a baseline for multilingual semantic retrieval but achieves limited performance, indicating that generic embeddings are insufficient for cross-lingual idiom understanding.

The E5 model improves performance under zero-shot settings, demonstrating the benefit of instruction-aware multilingual representations. However, the improvement remains modest, suggesting that pretrained embeddings alone are not sufficient for precise idiom retrieval.

The strongest results are obtained with the fine-tuned E5 model, which achieves 0.578 Top-1 and 0.760 Top-5 accuracy. This represents a substantial improvement over both baselines and highlights the importance of task-specific contrastive fine-tuning.

Overall, the results indicate that cross-lingual idiom retrieval requires not only strong multilingual representations but also explicit supervision to align Arabic contextual meaning with canonical idiomatic expressions.

Figure 19 illustrates the performance comparison across models, showing consistent gains at both Top-1 and Top-5.

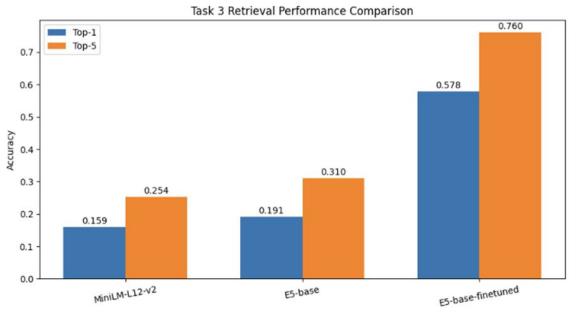


Figure 19: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Ablation Study and Model Progression*

The ablation study reveals a clear and consistent improvement across the retrieval pipeline, highlighting the impact of model architecture and training strategy.

As shown in Table 8, the MiniLM [28] baseline achieves limited performance, indicating that generic multilingual embeddings are insufficient for capturing idiomatic semantics in cross-lingual settings. Replacing MiniLM [28] with zero-shot multilingual E5 yields only modest gains, suggesting that pretrained multilingual representations alone are not sufficient for precise idiom alignment.

In contrast, fine-tuning the E5 model produces a substantial improvement, increasing Top-1 accuracy from 0.191 to 0.578 and Top-5 accuracy to 0.760. This represents a large relative gain over both baseline models.

These results confirm that task-specific contrastive training is the primary factor driving retrieval performance. Additional improvements are obtained through enhanced training strategies, including richer supervision and improved positive pair diversity, which contribute to better semantic alignment.

Overall, the ablation study demonstrates that idiom retrieval is not a general semantic similarity task, but a specialized problem that requires targeted supervision and fine-grained representation learning.

7. *Error Distribution Analysis*

To better understand model behavior, we categorize predictions into three outcome types:

- Top-1 correct: 6,662 queries (57.8%)
- Retrieved but misranked: 2,106 queries (18.3%)
- Fully incorrect: 2,762 queries (23.9%)

Figure 20 shows the distribution of these outcomes.

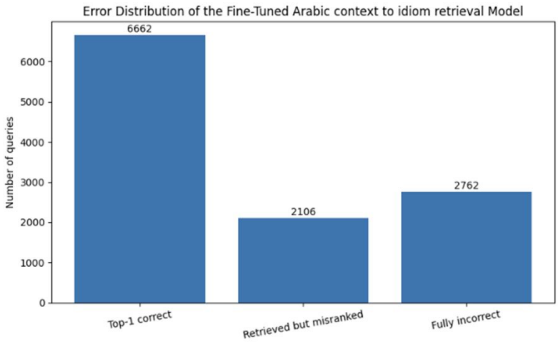


Figure 20: Error distribution across prediction outcomes.

A key observation is that a substantial portion of errors arises from ranking failures rather than retrieval failures. In many cases, the correct idiom is retrieved within the Top-5 candidates but is not ranked at the top position.

This indicates that the model successfully captures the overall semantic space but struggles with fine-grained discrimination between semantically similar idioms.

Fully incorrect predictions are typically associated with ambiguous Arabic inputs, weak contextual signals, or inherently difficult cross-lingual mappings.

Overall, the results suggest that improving ranking precision, rather than expanding retrieval coverage, is the primary direction for further performance gains.

8. *Similarity and Confidence Analysis*

We analyze the relationship between cosine similarity [5] scores and prediction correctness to assess model behavior and calibration.

Figure 21 shows that accuracy increases consistently with similarity score, indicating a strong positive correlation between similarity and prediction correctness. High-confidence predictions (similarity > 0.9) achieve very high accuracy, while low-similarity predictions are significantly less reliable.

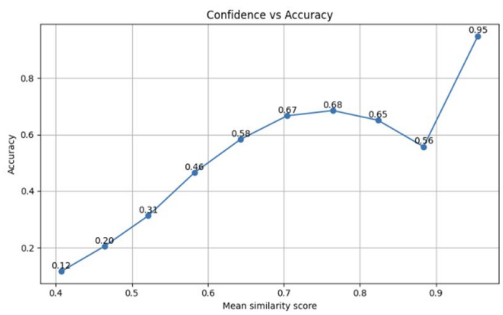


Figure 21: Accuracy as a function of similarity score.

Figure 22 further illustrates the distribution of similarity scores for correct and incorrect predictions. Correct predictions tend to have higher similarity values, while incorrect predictions are concentrated in lower score regions. However, there is a noticeable overlap between the two distributions.

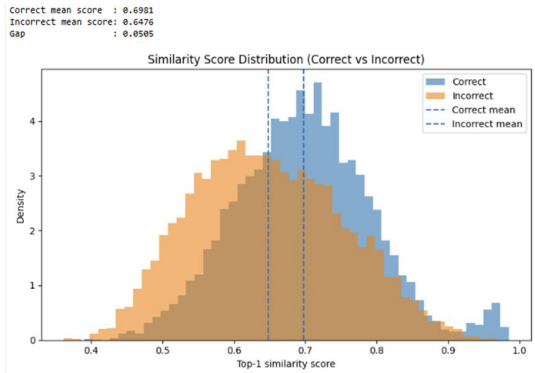


Figure 22: Similarity score distribution for correct and incorrect predictions.

This relatively small separation is also reflected in the mean similarity scores (0.698 for correct vs. 0.648 for incorrect), indicating that the embedding space is globally well-structured but locally dense.

The observed overlap suggests that many errors occur within semantically similar idiom clusters, rather than from complete semantic mismatches.

Overall, these findings show that cosine similarity provides a meaningful confidence signal but also highlight limitations in discriminative power when distinguishing between closely related idiomatic expressions.

9. *Threshold and Calibration Analysis*

We analyse model performance under varying similarity thresholds to evaluate its effectiveness as a confidence-based filtering mechanism.

Figure 23 shows that accuracy increases consistently as the similarity threshold becomes more restrictive. At lower thresholds, accuracy remains moderate due to the inclusion of ambiguous predictions, while higher thresholds (≥ 0.90) yield substantially higher accuracy, approaching near-perfect precision.

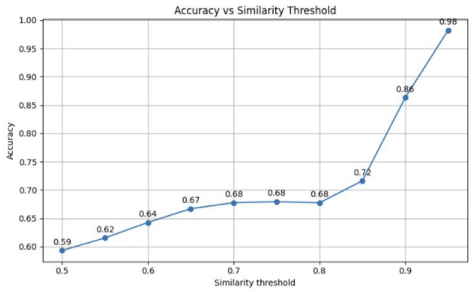


Figure 23: Accuracy as a function of similarity threshold.

This behavior reflects a clear trade-off between accuracy and coverage: stricter thresholds improve prediction reliability but reduce the number of accepted predictions.

These results indicate that similarity-based thresholding provides an effective calibration mechanism for practical deployment. By selecting an appropriate threshold, the system can balance precision and coverage, enable high-confidence predictions while deferring uncertain cases to fallback strategies.

10. *Recall@K Analysis*

We evaluate retrieval performance across different values of K to assess how often the correct idiom appears within the top-K candidates.

Figure 24 shows that recall increases rapidly for small values of K, with a large proportion of correct idioms retrieved within the Top-5 results. As K increases, the curve gradually saturates, indicating diminishing returns from retrieving additional candidates.

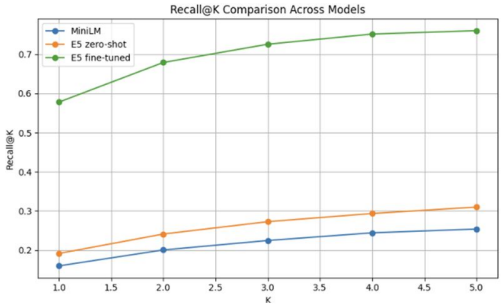


Figure 24: Recall@K performance across models.

This behavior suggests that the model is generally successful at identifying the correct semantic neighborhood, even when the correct idiom is not ranked at the top position.

In particular, the strong Recall@5 performance (0.760) indicates that the correct idiom is retrieved in most cases. However, the gap between Recall@1 and Recall@5 highlights persistent ranking limitations.

These findings reinforce earlier observations from the error analysis, confirming that the primary challenge lies not in retrieving relevant candidates, but in accurately ordering them within dense semantic clusters.

11. *Score Distribution Analysis*

Figure 25 compares the similarity score distributions across models. The fine-tuned E5 model produces a clear shift toward higher similarity values, indicating improved embedding alignment and stronger ranking confidence.

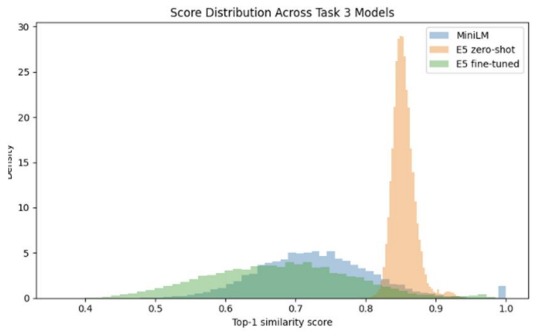


Figure 25: Similarity score distribution across models.

Compared to baseline models, the fine-tuned model exhibits better concentration of scores in higher similarity regions, reflecting improved semantic matching between Arabic contexts and idiomatic expressions.

This shift in distribution further confirms the effectiveness of task-specific fine-tuning in shaping a more discriminative embedding space.

12. *Qualitative Cross-Lingual Analysis*

To better understand model behavior, we present representative examples illustrating both successful predictions and typical failure cases.

The model performs well when the Arabic context provides clear semantic signals. For example:

- Arabic: "انتهى كل شيء في النهاية"
  Prediction: *come to an end*
- Arabic: "شارك همومك مع الآخرين لتخفيفها"
  Prediction: *a problem shared is a problem halved*
- Arabic: "لا توجد أي فرصة لحدوث ذلك"
  Prediction: *not a chance*

In these cases, the model successfully captures the underlying meaning and maps it to the appropriate idiomatic expression.

However, failure cases reveal consistent challenges:

- Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
  Prediction: *please speak more slowly*
  Target: *slow down*

Here, the model retrieves a semantically related phrase rather than the canonical idiom, indicating sensitivity to surface-level similarity.

Across examples, successful predictions are typically associated with strong contextual signals and clear semantic intent, while errors often arise in ambiguous contexts or when multiple idioms share similar meanings.

Notably, incorrect predictions are often assigned high similarity scores, indicating overconfidence in dense semantic regions. This behavior aligns with earlier findings and highlights limitations in fine-grained ranking.

13. *Hard* Negative Analysis

Analysis of hard negative cases reveals that incorrect predictions are frequently semantically close alternatives to the correct idiom.

For example, idioms such as *in one fell swoop* vs. *at one stroke*, or *packed to the rafters* vs. *stuffed to the gills*, often appear within the same candidate set and are difficult to distinguish.

This indicates that the model struggles to differentiate between idioms within the same semantic cluster, where distinctions are subtle and context-dependent.

These findings suggest that incorporating hard negative mining during training could improve ranking precision by encouraging the model to better separate closely related idiomatic expressions.

14. *Discussion*

The results demonstrate that Arabic context-to-idiom retrieval is significantly more challenging than its monolingual counterpart due to cross-lingual semantic gaps and variation in idiomatic expression.

Unlike generation-based approaches, retrieval models rely on the structure of the embedding space to align meanings across languages. While multilingual models improve this alignment, they remain sensitive to subtle semantic differences and cultural variations in idiomatic usage.

The experimental results show that replacing the MiniLM [28] baseline with multilingual E5 leads to consistent improvements in retrieval performance. However, the gains from pretrained multilingual representations alone remain limited. The most substantial improvement is achieved through task-specific contrastive fine-tuning, which enables more precise alignment between Arabic contextual inputs and canonical English idioms.

Despite strong overall performance, the model continues to struggle with fine-grained semantic distinctions, particularly among idioms within the same semantic cluster. This highlights an inherent limitation of purely retrieval-based approaches for idiom understanding.

These findings suggest that future work may benefit from hybrid or generative approaches that can better capture subtle semantic nuances beyond nearest-neighbor retrieval.

15. Task Summary

In this task, we introduced a cross-lingual context-to-idiom retrieval benchmark, requiring the mapping of Arabic contextual inputs to canonical English idioms within a fixed candidate inventory.

The fine-tuned multilingual E5 model achieves the best performance, demonstrating strong capability in capturing cross-lingual semantic alignment. However, detailed analysis shows that most remaining errors arise from ranking ambiguity rather than retrieval failure.

This indicates that the primary challenge lies in distinguishing between semantically similar idioms within dense embedding regions.

Overall, the results highlight that effective cross-lingual idiom retrieval requires not only strong multilingual representations but also targeted training strategies that improve ranking precision.

These findings position cross-lingual idiom retrieval as a challenging and promising direction for multilingual semantic modeling, with clear opportunities for improvement through ranking optimization, hard negative mining, and enhanced contextual modeling.

These results highlight that cross-lingual idiom retrieval is not only a retrieval problem but also a semantic alignment challenge, where models must bridge figurative meaning across languages rather than rely on surface similarity.

VIII. Discussion

The experimental results demonstrate that IdiomX supports a diverse range of NLP tasks related to idiom understanding, each requiring different modeling capabilities.

The findings highlight three key insights. First, idiom detection relies heavily on contextual modeling rather than surface-level lexical cues, confirming that idiomatic meaning cannot be inferred from isolated tokens. Second, context-to-idiom mapping is not purely a retrieval or generation problem, but requires precise semantic alignment between contextual meaning and canonical expressions. Third, hybrid approaches consistently outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

Across tasks, a consistent pattern emerges: while retrieval-based models are effective at identifying the correct semantic neighborhood, they often struggle with fine-grained ranking among semantically similar idioms. In contrast, generative models offer greater flexibility but may lack precision in producing canonical idiomatic forms. This suggests that idiom understanding benefits from a division of labor, where generative models provide contextual reasoning and retrieval mechanisms ensure discrete and accurate grounding.

In the cross-lingual setting (Task 3), these challenges become more pronounced due to semantic variation across languages. Although multilingual embeddings enable alignment between Arabic contexts and English idioms, performance remains limited by ambiguity within dense semantic clusters. This highlights the importance of improved ranking strategies and training methods that better capture subtle semantic distinctions.

Overall, the results position idiom understanding as a complex semantic task that requires integrating contextual reasoning, structured retrieval, and cross-lingual alignment within a unified framework.

IX. Applications

The IdiomX dataset supports a wide range of natural language processing tasks related to figurative language understanding, including:

- idiom detection
- context-to-idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly well-suited for evaluating transformer-based models and large language models on non-compositional language, where meaning cannot be derived directly from surface forms.

X. Limitations

While IdiomX provides large-scale coverage and rich annotations, several limitations should be considered.

First, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Second, the Arabic data primarily reflects standard written Arabic and may not fully capture dialectal variation.

In addition, some generated examples may reflect model-preferred phrasing rather than naturally occurring usage frequencies. For the generation task, evaluation relies primarily on exact match metrics, which may not fully capture semantic equivalence between idiomatic expressions.

Future work includes incorporating human validation and extending the dataset to additional languages and dialects.

XI. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing broad coverage, rich semantic annotations, and aligned English–Arabic representations.

Through comprehensive benchmarking, we showed that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches for idiom detection, while hybrid generation–retrieval methods achieve the best performance for context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, retrieval-based models are effective at capturing general semantic

alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

Overall, the results demonstrate that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

Future work includes extending IdiomX to additional languages using the same pipeline.

XII. Reproducibility and Public Resources

To ensure transparency and reproducibility, we release all resources associated with IdiomX, including the dataset, codebase, trained models, and interactive demonstrations.

Dataset

The IdiomX dataset is publicly available on Hugging Face:

- https://huggingface.co/datasets/aymansharara/IdiomX

It includes all configurations, splits, and enriched fields used in this work, supporting multiple tasks such as idiom detection and retrieval.

Code and Pipelines

The full dataset construction pipeline and experimental implementations are available on GitHub:

- Dataset pipeline: https://github.com/aymanshar/idiomx-dataset
- Models and benchmarks: https://github.com/aymanshar/IdiomX

These repositories include data processing scripts, training pipelines, and evaluation code to fully reproduce the experiments presented in this paper.

Interactive Demonstrations

We provide interactive demos for each core task via Hugging Face Spaces:

- Idiom Detection (Task 1): https://huggingface.co/spaces/aymansharara/idiomX_idiom_detection_demo
- Context → Idiom Retrieval (Task 2): https://huggingface.co/spaces/aymansharara/idiomx_context_to_idiom_demo
- Arabic Context → Idiom Retrieval (Task 3): https://huggingface.co/spaces/aymansharara/idiomx_arabic_context_to_idiom_demo

These demos allow users to explore model behavior interactively and evaluate predictions qualitatively.

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